Tejaswi Venumadhav Nerella — Curriculum Vitae

Assistant Professor, Department of Physics
Broida Hall
University of California, Santa Barbara
Santa Barbara, CA 93106-9530

Contact:
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email: teja@ucsb.edu

Education

California Institute of Technology 2010-2015
Ph.D. in Physics,
Advisor: Christopher Hirata

Indian Institute of Technology, Kanpur 2005-2010
M.Sc (Integrated) in Physics

Academic Honors

Sloan Research Fellowship 2023
Alfred P. Sloan Foundation

Hellman Family Faculty Fellowship 2023
Society of Hellman Fellows

John Bahcall Fellowship 2019
Institute for Advanced Study

Schmidt Fellowship 2015 - 2018
Institute for Advanced Study

Robert A. Millikan Fellowship 2010
California Institute of Technology

International Fulbright Science and Technology Award 2010
Bureau of Education and Cultural Affairs, U.S. Department of State

President's Gold Medal for the best academic performance in the graduating class in all disciplines, IIT Kanpur 2010

General Proficiency Medal for the best academic performance in the graduating class in Physics, IIT Kanpur 2010

Summer Undergraduate Research Fellowship 2007, 2008
California Institute of Technology

Academic Excellence Award 2007, 2008, 2009, 2010
IIT Kanpur

Silver Medal, 36th International Physics Olympiad 2005

KVPY Fellowship 2004
Department of Science and Technology, Govt. of India

NTSE Fellowship 2003
National Council of Educational Research and Training, Govt. of India

Work Experience

Assistant Professor Jul 2020-present
University of California, Santa Barbara

Visiting Professor 2020-present
International Center for Theoretical Sciences, Bangalore

Member Sep 2015-Jul 2020
Institute for Advanced Study, Princeton

Associate 2019-2020
International Center for Theoretical Sciences, Bangalore

Graduate Student Sep 2010-Aug 2015
California Institute of Technology, Pasadena
Advisor: Christopher M. Hirata

Visiting Scientist May-August 2009
Max-Planck-Institut für Physik komplexer Systeme, Dresden
Advisor: Roderich Moessner

Summer Undergraduate Research Fellow May-August 2008
California Institute of Technology, Pasadena
Advisor: Re'em Sari

Summer Undergraduate Research Fellow May-August 2007
California Institute of Technology, Pasadena
Advisor: Andrew Lange

Current and Pending Support

Project/Proposal Title:	WoU-MMA: Expanding the Horizons of Gravitational Wave Searches and Parameter Estimation
Source of Support:	NSF
Award Number:	2012086
Project Location:	University of California, Santa Barbara
Total Amount:	$300000
Starting/Ending Dates:	07/01/2020–06/30/2024
Person-months Per Year:	4 calendar months/year

Project/Proposal Title:	WoU-MMA: Targeted Search for Binary Mergers with Multiple Harmonics in Gravitational Wave Data
Source of Support:	NSF
Award Number:	2309360
Project Location:	University of California, Santa Barbara
Total Amount:	$300000
Starting/Ending Dates:	05/01/2023–04/30/2026
Person-months Per Year:	4 calendar months/year

Project/Proposal Title:	Searching for Missing Links in the Black Hole Population
Source of Support:	Alfred P. Sloan Foundation
Project Location:	University of California, Santa Barbara
Total Amount:	$75000
Starting/Ending Dates:	09/15/2023–09/15/2025
Person-months Per Year:	2 calendar months/year

Project/Proposal Title:	Cosmological recombination on small-scales, and implications for the Hubble constant
Source of Support:	Society of Hellman Fellows
Project Location:	University of California, Santa Barbara
Total Amount:	$24800
Starting/Ending Dates:	06/01/2023–08/31/2024
Person-months Per Year:	2 calendar months/year

Project/Proposal Title:	IAS Gravitational Wave Virtual Center
Source of Support:	Jonathan M. Nelson Center for Collaborative Research
Project Location:	Institute for Advanced Study
Total Amount:	$300000
Starting/Ending Dates:	07/01/2026–07/01/2029
Person-months Per Year:	3 (Co-Investigator, travel support, and partial postdoc support)
URL:	`https://www.ias.edu/nccr/projects`

Graduate Committees

Student Name	Degree Completed	Role
Anantpurkar, Isha	In progress	Chair
Schiff, Jonathan	In progress	Chair
Garza, Sierra	In progress	Chair
Kwon, James (Kyubin)	In progress	Chair
Becerra Espinoza, Joaquin	In progress	Chair
Surgent, William	In progress	Chair
Huang, Jiamu	In progress	Member
Gonzalez-Hernandez, Diego	In progress	Member
Clay, Hawkins	In progress	Member
Kim, Crystal	In progress	Member
Breckenridge, Josh	In progress	Member
Vancil, Connor	In progress	Member
Palatnick, Skyler	In progress	Member
Liang, Naixin	In progress	Member

Undergraduate Students Supervised

Name	Years	Co-Mentor	Project Title	Position After Project
Liu, Cuishan	2021–2022	Javier Roulet (KITP)	Coordinates for Gravitational Wave Parameter Estimation using Fisher Matrices	Graduate Student (University of Virginia)
Carrel, Dashiell	2021–2023	Jennifer Barnes (KITP)	AT2017gfo-like Photometry From a Single Component of Binary Neutron Star Merger Ejecta	Graduate Student (University of California Berkeley)
Wang, Jiayu	2022–2024	Ajit Kumar Mehta (UCSB postdoc)	Improving the Detection Sensitivity of Gravitational-Wave Searches in Non-Stationary Data	

Wei, Fansen	2022–2023		Calibration Uncertainty in Gravitational-Wave Parameter Estimation	Graduate Student (Brown University)
Xu, Junyan	2023–2025		Parameter Estimation of Long-Duration Signals in Third-Generation Gravitational Wave Detectors	Graduate Student (University of Minnesota)
Ding, Yangxiao	2025–Present	James Kwon	Interface modes in neutron stars	
Hahn, Christian	2026–Present	Tousif Islam	TBD	

Postdoctoral Scholars Supervised

Name	Years	Note	Position After UCSB
Roulet, Javier	2021–2022	KITP Fellow	Sherman Fairchild Postdoctoral Scholar (Caltech)
Barnes, Jennifer	2021-2023	KITP Fellow	Senior Algorithm Engineer, 3M, Minneapolis, Minnesota
Yu, Hang	2022-2023	KITP Fellow	Assistant Professor, Montana State University
Ajit Kumar Mehta	2022-2025		Assistant Professor, Chennai Mathematical Institute
Tousif Islam	2023-Present	KITP Fellow	

Other Supervision/Mentoring

Name	Years	Note	Position After Supervision
Islam, Tousif	2022–2023	KITP Graduate Fellow	KITP Fellow

University Service

Name of Committee/Organization	Period	Position
Graduate Admissions, Physics Department	2021-2023	Member
Winter quarter Colloquium Committee, Physics Department	2021-2022	Co-Chair
Spring quarter Colloquium Committee, Physics Department	2022-2023	Co-Chair
Student Awards, Physics Department	2024-2025	Chair
Freshman Advisor, Physics Department	2025-2026	Advisor
Undergraduate Research, Physics Department	2025-2026	Member

Professional Service

- Referee for the Astrophysical Journal Letters
- Referee for Astroparticle Physics
- Referee for the Astrophysical Journal
- Referee for Monthly Notices of the Royal Astronomical Society Letters
- Referee for Monthly Notices of the Royal Astronomical Society
- Referee for Physical Review D
- Panel reviewer for ERC, BSF, NSF
- Organizer of Prospects in Theoretical Physics 2025 program on *Gravitational Waves from Theory to Observation URL:* *`https://www.ias.edu/pitp-2025-details`*
- *Organizer of the 2025 KITP Program:* Stellar-Mass Black Holes at the Nexus of Optical, X-ray, and Gravitational Wave Surveys URL: `https://www.kitp.ucsb.edu/activities/stellarbh25`
- Organizer of the 2025 KITP Conference: *The Lifecycle of Stellar Black Holes URL:* *`https://www.kitp.ucsb.edu/activities/stellarbh-c25`*

Publication Highlights

Total h-index: 32

Top five cited published papers (not counting n^{th} author papers):

1. **Venumadhav, T.**, Zackay, B., Roulet, J., Dai, L., Zaldarriaga, M., (2020), Physical Review D, 101, 083030
   Title: New binary black hole mergers in the second observing run of Advanced LIGO and Advanced Virgo
   Citation count: 378

2. Olsen, S., **Venumadhav, T.**, Mushkin, J., Roulet, J., Zackay, B., Zaldarriaga, M., (2022), Physical Review D, 106, 043009
   Title: New binary black hole mergers in the LIGO-Virgo O3a data
   Citation count: 204

3. Zackay, B., **Venumadhav, T.**, Dai, L., Roulet, J., Zaldarriaga, M., (2019), Physical Review D, 100, 023007
   Title: Highly spinning and aligned binary black hole merger in the Advanced LIGO first observing run
   Citation count: 200

4. **Venumadhav, T.**, Zackay, B., Roulet, J., Dai, L., Zaldarriaga, M., (2019), Physical Review D, 100, 023011
   Title: New search pipeline for compact binary mergers: Results for binary black holes in the first observing run of Advanced LIGO
   Citation count: 188

5. Zackay, B., Dai, L., **Venumadhav, T.**, Roulet, J., Zaldarriaga, M., (2021), Physical Review D, 104, 063030
   Title: Detecting gravitational waves with disparate detector responses: Two new binary black hole

mergers
Citation count: 155

Refereed Publications

1. Islam, T., **Venumadhav, T.**, Wadekar, D., (2026), Physical Review Letters, 137, 031404
   Title: Progenitor of the Recoiling Supermassive Black Hole RBH-1 Identified Using HST and JWST Imaging
2. Maity, K.N., Jana, S., **Venumadhav, T.**, Barsode, A., Ajith, P., (2026), Physical Review D, 113, 103522
   Title: Strong lensing cosmography using binary-black-hole mergers: Prospects for the near future
3. Hegade K. R., A., Kwon, K.J., **Venumadhav, T.**, Yu, H., Yunes, N., (2026), Physical Review Letters, 136, 071401
   Title: Relativistic and Dynamical Love Numbers
4. Cheung, M.H.Y., Wadekar, D., Mehta, A.K., Islam, T., Roulet, J., Berti, E., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2026), Physical Review D, 113, 023003
   Title: Searching for intermediate mass ratio binary black hole mergers in the third observing run of LIGO-Virgo-KAGRA
5. Mehta, A.K., Wadekar, D., Anantpurkar, I., Roulet, J., **Venumadhav, T.**, Islam, T., Mushkin, J., Zackay, B., Zaldarriaga, M., (2025), Physical Review D, 112, 124023
   Title: Binary black hole population inference combining confident and marginal events from the IAS-HM search pipeline
6. Mushkin, J., Roulet, J., Zackay, B., **Venumadhav, T.**, Ivashtenko, O., Wadekar, D., Mehta, A.K., Zaldarriaga, M., (2025), Physical Review D, 112, 104025
   Title: Sampler-free gravitational wave inference using matrix multiplication
7. Perera, A., Zackay, B., **Venumadhav, T.**, (2025), The Astrophysical Journal Supplement Series, 281, 4
   Title: A New Search Pipeline for Short Gamma-Ray Bursts in Fermi/GBM Data—A 50% Increase in the Number of Detections
8. Islam, T., **Venumadhav, T.**, (2025), Physical Review D, 112, 104039
   Title: Post-Newtonian theory-inspired framework for characterizing eccentricity in gravitational waveforms
9. Jana, S., Kapadia, S.J., **Venumadhav, T.**, More, S., Ajith, P., (2025), Physical Review Letters, 135, 111402
   Title: Probing the Nature of Dark Matter Using Strongly Lensed Gravitational Waves from Binary Black Holes
10. Islam, T., **Venumadhav, T.**, Mehta, A.K., Anantpurkar, I., Wadekar, D., Roulet, J., Mushkin, J., Zackay, B., Zaldarriaga, M., (2025), Physical Review D, 112, 044070
    Title: Data-driven extraction, phenomenology, and modeling of eccentric harmonics in binary black hole merger waveforms
11. Islam, T., **Venumadhav, T.**, (2025), Physical Review D, 111, L081503
    Title: Universal phenomenological relations between spherical harmonic modes in nonprecessing eccentric binary black hole merger waveforms
12. Mehta, A.K., Olsen, S., Wadekar, D., Roulet, J., **Venumadhav, T.**, Mushkin, J., Zackay, B., Zaldarriaga, M., (2025), Physical Review D, 111, 024049
    Title: New binary black hole mergers in the LIGO-Virgo O3b data
13. Jana, S., J Kapadia, S., **Venumadhav, T.**, More, S., Ajith, P., (2024), Classical and Quantum Gravity, 41, 245010
    Title: Strong-lensing cosmography using third-generation gravitational-wave detectors

14. Wadekar, D., **Venumadhav, T.**, Mehta, A.K., Roulet, J., Olsen, S., Mushkin, J., Zackay, B., Zaldarriaga, M., (2024), Physical Review D, 110, 084035
Title: New approach to template banks of gravitational waves with higher harmonics: Reducing matched-filtering cost by over an order of magnitude

15. Chia, H.S., Edwards, T.D.P., Wadekar, D., Zimmerman, A., Olsen, S., Roulet, J., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2024), Physical Review D, 110, 063007
Title: In pursuit of Love numbers: First templated search for compact objects with large tidal deformabilities in the LIGO-Virgo data

16. Wadekar, D., **Venumadhav, T.**, Roulet, J., Mehta, A.K., Zackay, B., Mushkin, J., Zaldarriaga, M., (2024), Physical Review D, 110, 044063
Title: New search pipeline for gravitational waves with higher-order modes using mode-by-mode filtering

17. Roulet, J., Mushkin, J., Wadekar, D., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2024), Physical Review D, 110, 044010
Title: Fast marginalization algorithm for optimizing gravitational wave detection, parameter estimation, and sky localization

18. Roulet, J., **Venumadhav, T.**, (2024), Annual Review of Nuclear and Particle Science, 74, 207
Title: Inferring Binary Properties from Gravitational-Wave Signals

19. Yu, H., Roulet, J., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2023), Physical Review D, 108, 064059
Title: Accurate and efficient waveform model for precessing binary black holes

20. Jana, S., Kapadia, S.J., **Venumadhav, T.**, Ajith, P., (2023), Physical Review Letters, 130, 261401
Title: Cosmography Using Strongly Lensed Gravitational Waves from Binary Black Holes

21. Yu, H., Weinberg, N.N., Arras, P., Kwon, J., **Venumadhav, T.**, (2023), Monthly Notices of the Royal Astronomical Society, 519, 4325
Title: Beyond the linear tide: impact of the non-linear tidal response of neutron stars on gravitational waveforms from binary inspirals

22. Roulet, J., Olsen, S., Mushkin, J., Islam, T., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2022), Physical Review D, 106, 123015
Title: Removing degeneracy and multimodality in gravitational wave source parameters

23. Olsen, S., **Venumadhav, T.**, Mushkin, J., Roulet, J., Zackay, B., Zaldarriaga, M., (2022), Physical Review D, 106, 043009
Title: New binary black hole mergers in the LIGO-Virgo O3a data

24. Chia, H.S., Olsen, S., Roulet, J., Dai, L., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2022), Physical Review D, 106, 024009
Title: Signs of higher multipoles and orbital precession in GW151226

25. Roulet, J., Chia, H.S., Olsen, S., Dai, L., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2021), Physical Review D, 104, 083010
Title: Distribution of effective spins and masses of binary black holes from the LIGO and Virgo O1-O3a observing runs

26. Olsen, S., Roulet, J., Chia, H.S., Dai, L., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2021), Physical Review D, 104, 083036
Title: Mapping the likelihood of GW190521 with diverse mass and spin priors

27. Zackay, B., Dai, L., **Venumadhav, T.**, Roulet, J., Zaldarriaga, M., (2021), Physical Review D, 104, 063030
Title: Detecting gravitational waves with disparate detector responses: Two new binary black hole mergers

28. Zackay, B., **Venumadhav, T.**, Roulet, J., Dai, L., Zaldarriaga, M., (2021), Physical Review D, 104, 063034
Title: Detecting gravitational waves in data with non-stationary and non-Gaussian noise

29. Roulet, J., **Venumadhav, T.**, Zackay, B., Dai, L., Zaldarriaga, M., (2020), Physical Review D, 102, 123022
Title: Binary black hole mergers from LIGO/Virgo O1 and O2: Population inference combining confident and marginal events

30. Huang, Y., Haster, C.J., Roulet, J., Vitale, S., Zimmerman, A., **Venumadhav, T.**, Zackay, B., Dai, L., Zaldarriaga, M., (2020), Physical Review D, 102, 103024
Title: Source properties of the lowest signal-to-noise-ratio binary black hole detections

31. **Venumadhav, T.**, Zackay, B., Roulet, J., Dai, L., Zaldarriaga, M., (2020), Physical Review D, 101, 083030
Title: New binary black hole mergers in the second observing run of Advanced LIGO and Advanced Virgo

32. Dai, L., Kaurov, A.A., Sharon, K., Florian, M., Miralda-Escudé, J., **Venumadhav, T.**, Frye, B., Rigby, J.R., Bayliss, M., (2020), Monthly Notices of the Royal Astronomical Society, 495, 3192
Title: Asymmetric surface brightness structure of caustic crossing arc in SDSS J1226+2152: a case for dark matter substructure

33. Samsing, J., **Venumadhav, T.**, Dai, L., Martinez, I., Batta, A., Lopez, M., Ramirez-Ruiz, E., Kremer, K., (2019), Physical Review D, 100, 043009
Title: Probing the black hole merger history in clusters using stellar tidal disruptions

34. Kaurov, A.A., Dai, L., **Venumadhav, T.**, Miralda-Escudé, J., Frye, B., (2019), The Astrophysical Journal, 880, 58
Title: Highly Magnified Stars in Lensing Clusters: New Evidence in a Galaxy Lensed by MACS J0416.1-2403

35. **Venumadhav, T.**, Zackay, B., Roulet, J., Dai, L., Zaldarriaga, M., (2019), Physical Review D, 100, 023011
Title: New search pipeline for compact binary mergers: Results for binary black holes in the first observing run of Advanced LIGO

36. Zackay, B., **Venumadhav, T.**, Dai, L., Roulet, J., Zaldarriaga, M., (2019), Physical Review D, 100, 023007
Title: Highly spinning and aligned binary black hole merger in the Advanced LIGO first observing run

37. Roulet, J., Dai, L., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2019), Physical Review D, 99, 123022
Title: Template bank for compact binary coalescence searches in gravitational wave data: A general geometric placement algorithm

38. Dai, L., **Venumadhav, T.**, Kaurov, A.A., Miralda-Escud, J., (2018), The Astrophysical Journal, 867, 24
Title: Probing Dark Matter Subhalos in Galaxy Clusters Using Highly Magnified Stars

39. **Venumadhav, T.**, Dai, L., Kaurov, A., Zaldarriaga, M., (2018), Physical Review D, 98, 103513
Title: Heating of the intergalactic medium by the cosmic microwave background during cosmic dawn

40. Kaurov, A.A., **Venumadhav, T.**, Dai, L., Zaldarriaga, M., (2018), The Astrophysical Journal, 864, L15
Title: Implication of the Shape of the EDGES Signal for the 21 cm Power Spectrum

41. Hirata, C.M., Mishra, A., **Venumadhav, T.**, (2018), Physical Review D, 97, 103521
Title: Detecting primordial gravitational waves with circular polarization of the redshifted 21 cm line. I. Formalism

42. **Venumadhav, T.**, Dai, L., Miralda-Escudé, J., (2017), The Astrophysical Journal, 850, 49
Title: Microlensing of Extremely Magnified Stars near Caustics of Galaxy Clusters

43. **Venumadhav, T.**, Oklopčić, A., Gluscevic, V., Mishra, A., Hirata, C.M., (2017), Physical Review D, 95, 083010
Title: New probe of magnetic fields in the prereionization epoch. I. Formalism

44. Gluscevic, V., **Venumadhav, T.**, Fang, X., Hirata, C., Oklopčić, A., Mishra, A., (2017), Physical Review D, 95, 083011
Title: New probe of magnetic fields in the pre-reionization epoch. II. Detectability
45. Dai, L., **Venumadhav, T.**, Sigurdson, K., (2017), Physical Review D, 95, 044011
Title: Effect of lensing magnification on the apparent distribution of black hole mergers
46. **Venumadhav, T.**, Chang, T.C., Doré, O., Hirata, C.M., (2016), The Astrophysical Journal, 826, 116
Title: A Practical Theorem on Using Interferometry to Measure the Global 21-cm Signal
47. **Venumadhav, T.**, Cyr-Racine, F.Y., Abazajian, K.N., Hirata, C.M., (2016), Physical Review D, 94, 043515
Title: Sterile neutrino dark matter: Weak interactions in the strong coupling epoch
48. **Venumadhav, T.**, Hirata, C., (2015), Physical Review D, 91, 123009
Title: Stability of small-scale baryon perturbations during cosmological recombination
49. **Venumadhav, T.**, Zimmerman, A., Hirata, C.M., (2014), The Astrophysical Journal, 781, 23
Title: The Stability of Tidally Deformed Neutron Stars to Three- and Four-mode Coupling
50. Venumadhav, T., Haque, M., Moessner, R., (2010), Physical Review B, 81, 054305
Title: Finite-rate quenches of site bias in the Bose-Hubbard dimer

Preprints on the Arxiv

1. Yu, H., Nicolini, G., Lau, S.Y., Kwon, K.J., **Venumadhav, T.**, Andersson, N., Pnigouras, P., Gittins, F., Nanda, A., (2026), arXiv:2607.07943
Title: Nonlinear hydrodynamics in spinning neutron stars: Theoretical universal relations and equilibrium solutions
2. Abhishek Hegade K., R., Kwon, K.J., **Venumadhav, T.**, Yu, H., Yunes, N., (2026), arXiv:2605.08569
Title: The Good, the Bad, and the Subtle: Relativistic mode sums for neutron-star tidal response
3. Perera, A., Zackay, B., **Venumadhav, T.**, (2026), arXiv:2605.31554
Title: Expanding the Population of Short Gamma-Ray Transients with a Coherent Fermi/GBM Search. A 13-year catalog of short GRBs
4. Islam, T., Wadekar, D., **Venumadhav, T.**, Zaldarriaga, M., Mehta, A.K., Roulet, J., Zackay, B., (2026), arXiv:2605.11280
Title: Discovery of Interpretable Surrogates via Agentic AI: Application to Gravitational Waves
5. Islam, T., **Venumadhav, T.**, Wadekar, D., Mehta, A.K., Roulet, J., Mushkin, J., Ho-Yeuk Cheung, M., Zackay, B., Zaldarriaga, M., (2026), arXiv:2604.07388
Title: GW190711_030756 and GW200114_020818: astrophysical interpretation of two asymmetric binary black hole mergers in the IAS catalog
6. Zhou, Z., Wadekar, D., Roulet, J., Ivashtenko, O., **Venumadhav, T.**, Islam, T., Mehta, A.K., Mushkin, J., Ho-Yeuk Cheung, M., Zackay, B., Zaldarriaga, M., (2026), arXiv:2603.05784
Title: Searching for precessing binary systems with mode-by-mode filtering and marginalization
7. Islam, T., **Venumadhav, T.**, Mehta, A.K., Wadekar, D., Roulet, J., Anantpurkar, I., Mushkin, J., Zackay, B., Zaldarriaga, M., (2025), arXiv:2509.20556
Title: Data-driven extraction and phenomenology of eccentric harmonics in eccentric spinning binary black hole mergers
8. Wadekar, D., Pimpalkar, A., Ho-Yeuk Cheung, M., Wandelt, B., Berti, E., Mehta, A.K., **Venumadhav, T.**, Roulet, J., Islam, T., Zackay, B., Mushkin, J., Zaldarriaga, M., (2025), arXiv:2507.08318
Title: Improving gravitational wave search sensitivity with TIER: Trigger Inference using Extended strain Representation
9. Schiff, J., **Venumadhav, T.**, (2025), arXiv:2506.16517
Title: Primordial magnetic fields and modified recombination histories

10. Islam, T., **Venumadhav, T.**, Mehta, A.K., Anantpurkar, I., Wadekar, D., Roulet, J., Mushkin, J., Zackay, B., Zaldarriaga, M., (2025), arXiv:2504.12420
    Title: gwharmone: first data-driven surrogate for eccentric harmonics in binary black hole merger waveforms
11. Kwon, K.J., Yu, H., **Venumadhav, T.**, (2025), arXiv:2503.11837
    Title: Resonance locking: radian-level phase shifts due to nonlinear hydrodynamics of g-modes in merging neutron star binaries
12. Mehta, A.K., Wadekar, D., Roulet, J., Anantpurkar, I., **Venumadhav, T.**, Mushkin, J., Zackay, B., Zaldarriaga, M., Islam, T., (2025), arXiv:2501.17939
    Title: Significant increase in sensitive volume of a gravitational wave search upon including higher harmonics
13. Kwon, K.J., Yu, H., **Venumadhav, T.**, (2024), arXiv:2410.03831
    Title: Resonance Locking of Anharmonic g-Modes in Coalescing Neutron Star Binaries
14. Wadekar, D., Roulet, J., **Venumadhav, T.**, Mehta, A.K., Zackay, B., Mushkin, J., Olsen, S., Zaldarriaga, M., (2023), arXiv:2312.06631
    Title: New black hole mergers in the LIGO-Virgo O3 data from a gravitational wave search including higher-order harmonics
15. Islam, T., Roulet, J., **Venumadhav, T.**, (2022), arXiv:2210.16278
    Title: Factorized Parameter Estimation for Real-Time Gravitational Wave Inference
16. Dai, L., Zackay, B., **Venumadhav, T.**, Roulet, J., Zaldarriaga, M., (2020), arXiv:2007.12709
    Title: Search for Lensed Gravitational Waves Including Morse Phase Information: An Intriguing Candidate in O2
17. Coleman, M.S.B., **Venumadhav, T.**, Zackay, B., (2019), arXiv:1903.04978
    Title: Gravitational-wave-moderated Accretion: The Case of ES Ceti
18. Haris, K., Mehta, A.K., Kumar, S., **Venumadhav, T.**, Ajith, P., (2018), arXiv:1807.07062
    Title: Identifying strongly lensed gravitational wave signals from binary black hole mergers
19. Zackay, B., Dai, L., **Venumadhav, T.**, (2018), arXiv:1806.08792
    Title: Relative Binning and Fast Likelihood Evaluation for Gravitational Wave Parameter Estimation
20. Dai, L., **Venumadhav, T.**, Zackay, B., (2018), arXiv:1806.08793
    Title: Parameter Estimation for GW170817 using Relative Binning
21. Dai, L., **Venumadhav, T.**, (2017), arXiv:1702.04724
    Title: On the waveforms of gravitationally lensed gravitational waves

N-th Author Papers

1. Raaijmakers, G., et. al., (2021), The Astrophysical Journal, 922, 269
   Title: The Challenges Ahead for Multimessenger Analyses of Gravitational Waves and Kilonova: A Case Study on GW190425
2. Weltman, A., et. al., (2020), Publications of the Astronomical Society of Australia, 37, e002
   Title: Fundamental physics with the Square Kilometre Array
3. Doré, O., et. al., (2014), arXiv e-prints
   Title: Cosmology with the SPHEREX All-Sky Spectral Survey

Talks and Presentations

1. Invited theory colloquium, Tata Institute of Fundamental Research, Mumbai. 2025
2. Invited talk, Current Themes in Astrophysics and Particle Physics, NBIA, Copenhagen. 2025
3. Invited seminar, International Center for Theoretical Sciences, Bangalore. 2025
4. Invited review talk, APS Global Physics Summit 2025, Anaheim. 2025
5. Invited seminar, Weinberg Institute, University of Texas, Austin. 2025

6. Invited seminar, City University of Hong Kong, Hong Kong. 2025
7. Invited plenary talk, IAS Program on Fundamental Physics, HKUST, Hong Kong. 2025
8. Invited DGRAV seminar, APS DGRAV (online). 2024
9. Invited theory colloquium, Tata Institute of Fundamental Research, Mumbai. 2024
10. Invited talk, 21st Century Cosmology: tensions, anomaly and new physics, Ashoka University, Sonipat. 2024
11. Invited talk, XGMDC, Penn State University, PA. 2024
12. Invited seminar, IIT Hyderabad, Hyderabad. 2023
13. Invited talk, RESCEU-NBI Workshop on Gravitational-Wave Sources, Tokyo. 2023
14. Invited talk, International Conference on Gravitation and Cosmology (ICGC), Guwahati. 2023
15. Invited talk, Gravitational waves in the desert, Ein Gedi, Israel[1]. 2023
16. Invited talk, Gravitational Wave Physics and Astronomy Workshop, Melbourne. 2022
17. Invited Astrophysics seminar, International Center for Theoretical Sciences, Bangalore. 2022
18. Invited Astrophysics seminar, Raman Research Institute, Bangalore. 2022
19. Invited Astrophysics seminar, Indian Institute of Science, Bangalore. 2022
20. Invited talk at the ISSI (International Space Science Institute) Workshop on Strong Gravitational Lensing (virtual). 2022
21. Invited talk at the SRITP workshop on "EM counterparts to GW sources" at the Weizmann Institute, Rehovot. 2022
22. Invited talk at the KITP conference titled “Storming the Gravitational Wave Frontier". 2022
23. Invited Panelist at the APS April Meeting on the panel Data Analysis in Astrophysics (virtual). 2021
24. Invited Astrophysics Colloquium, Massachusetts Institute of Technology. 2020
25. Invited Cosmology seminar (virtual), CERN. 2020
26. Invited Seminar, CITA, Toronto. 2020
27. Invited Talk, Gravitational wave searches and parameter estimation in the era of detections, Schloss Ringberg. 2020
28. Invited Seminar, Indian Institute of Technology, Mumbai. 2020
29. Invited Seminar, Tata Institute of Fundamental Research, Mumbai. 2019
30. Invited Talk, Frank N. Bash Symposium, UT Austin. 2019
31. Talk, Gravitational Wave Physics and Astronomy Workshop, Tokyo. 2019
32. Invited Talk, Black Holes and Neutron Stars with Gravitational Waves, YITP, Kyoto. 2019
33. Invited Colloquium, Black Hole Initiative, Harvard. 2019
34. Invited panelist, The Future of Gravitational-Wave Astronomy, Bangalore. 2019
35. Invited Seminar, International Centre for Theoretical Sciences, TIFR. 2019
36. Invited Seminar, Princeton Gravity Initiative, Princeton. 2019
37. Invited Seminar, Albert Einstein Institute, Potsdam. 2019
38. Invited Seminar, Center for Cosmology and Particle Physics, NYU. 2019
39. Invited Seminar, Astronomy and Astrophysics, UC Santa Barbara. 2019
40. Invited Colloquium, Department of Physics, UC Santa Barbara. 2019
41. Invited panelist, Physics and Astrophysics at the eXtreme, IUCAA, Pune. 2018
42. Invited talk, Thermal history of the Universe at intermediate redshift: progress with 21cm absorption measurements, CERN. 2018

[1] Postponed

43. Talk, Shedding Light on the Dark Universe with Extremely Large Telescopes, UCLA. 2018
44. Invited Cosmology seminar, JHU, Baltimore. 2017
45. Invited Seminar, CITA, Toronto. 2017
46. Talk, Fundamental Physics with the Square Kilometer Array, Mauritius. 2017
47. Invited talk, Tianlai Collaboration Meeting, Fermilab, Batavia. 2016
48. Invited talk, CMB Spectral Distortions From Cosmic Baryon Evolution, RRI, Bengaluru. 2016
49. Invited seminar, International Centre for Theoretical Sciences, TIFR. 2016
50. Invited cosmology seminar, Perimeter institute. 2016
51. Cosmology lunch, joint w/ IAS and Princeton University. 2016
52. Astrophysics informal seminar, IAS. 2016
53. Seminar, Inter University Center for Astronomy and Astrophysics, Pune. 2015
54. Seminar, National Center for Radio Astronomy, Pune. 2015
55. Talk, The Primordial Universe after Planck, IAP, Paris. 2014
56. Seminar, McGill University, Montreal. 2014
57. Seminar, CITA, Toronto. 2014
58. ITC Seminar, Harvard University, Boston. 2014
59. Cosmology lunch, joint w/ IAS and Princeton University. 2014
60. Talk, Theoretical Astrophysics in Southern California (TASC), UCSD, San Diego. 2014
61. Special seminar, KICP, University of Chicago. 2014
62. Cosmology Lunch talk, CCAPP, Ohio State University, Columbus. 2014
63. Poster, Gravitational Wave Physics and Astronomy Workshop (GWPAW) at IUCAA, Pune. 2013
64. Seminar, Inter University Center for Astronomy and Astrophysics, Pune. 2013
65. Talk, Theoretical Astrophysics in Southern California (TASC), Carnegie Observatories, Pasadena. 2012
66. Poster, Summer school on cosmology, ICTP, Trieste. 2012

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